

**SAHAKAR MAHARSHI SHANKAR RAO MOHITE PATIL  
INSTITUTE OF TECHNOLOGY AND RESEARCH, AKLUJ  
DEPARTMENT OF COMPUTER ENGINEERING**

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**PROJECT SYNOPSIS TITLE**

**COLLEGE EVENT MANAGEMENT SYSTEM**

**NAMES OF GROUP MEMBERS:**

- 1. Shravani sanjay bankar:24213230010**
- 2. Sanika mahavir jadhav:24213230095**
- 3. Rutuja arjun thaware:24213230098**
- 4. Mahevish shabbir shaikh:24213230123**

## **1. INTRODUCTION & IMPORTANCE OF PROJECT AREA**

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The College Event Management System is a web-based platform designed to facilitate efficient planning, administration, and execution of various college events. Traditional event handling in academic institutions relies heavily on manual processing, physical forms, paper-bound record-keeping, and scattered coordination channels, leading to high administrative overhead and operational friction.

This system fundamentally reduces manual paperwork and optimizes record-keeping efforts by offering an integrated management workspace. It provides quick and highly accurate online student registration capabilities, improves active communication channels between event organizers and active participants, and guarantees complete data integrity by safely maintaining centralized storage. By automating scheduling, tracking, and reporting workflows, the application saves substantial time, reduces individual administrative burdens, automatically generates detailed analytical reports for smarter structural decision-making, and significantly upgrades the operational quality of handling events on campus.

## **2. PROJECT OBJECTIVES**

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The main architectural objectives guiding the development of this platform are defined below:

1. **Centralized Event Management:** Provide a unified console to configure, monitor, modify, and control all structural college events from a single operational dashboard.
2. **Easy Event Registration:** Enable swift end-user registration mechanics allowing students to safely enroll in chosen modules seamlessly online.
3. **Eliminate Manual Tasks:** Digitalize tracking structures to remove outdated paper-bound tracking and physical registration entries.
4. **Automated Event Scheduling:** Systematically create, manage, and distribute timelines, session updates, and explicit venue maps.
5. **Participant & Attendance Management:** Implement administrative tools to cleanly map incoming participant metrics, student sign-ups, and track actual attendance lists.
6. **Real-Time Notifications & Updates:** Keep the student base actively informed regarding unexpected date alterations, structural shifts, or upcoming deadlines.
7. **Automated Report Generation:** Produce detailed summaries analyzing overall participant registrations, final event attendance ratios, and performance metrics.
8. **Improved Inter-Communication:** Bridge operational gaps by building reliable direct interaction loops between core event managers and targeted participants.

### 3. SYSTEM DESIGN REQUIREMENTS

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#### 3.1 Input Design

- **Student Registration Details:** Collection of accurate profile configurations including full names, structural roll numbers, active contact links, and institutional credentials.
- **Secure Login Credentials:** Encrypted user validation metrics (Username/Passwords) to isolate unauthorized platform access layers.
- **Event Configuration Blocks:** Input forms mapping comprehensive event components including explicit title names, execution dates, allocated venues, and core descriptors.
- **Participant Registration Information:** Dynamic validation entry captures verifying specific event configurations chosen during student enrollment requests.

#### 3.2 Process Design

- **User Authentication (Login/Logout):** Structural validation layers handling session generation, authentication controls, and user termination procedures.

- **Event Creation and Management:** Core backend handling data insertion, data modifications, and standard record deletions regarding upcoming events.
- **Participant Registration Flow:** Algorithmic orchestration dealing with mapping specific entries, database validations, and updating capacity tracking.
- **Attendance Management:** Functional logic dealing with mapping active attendees, recording user logs, and calculating physical presentee ratios.
- **Report Generation:** Automated programmatic collection aggregating raw transactional data tables into readable quantitative documents.

### 3.3 Output Design

- **Dynamic Event Details Display:** Clean, responsive visual presentation layer showing active timelines and event descriptors.
- **Participant List:** Administrative reporting views showcasing comprehensive tables of registered students categorized per event.
- **Attendance Report:** Formatted summaries tracking live participant details against the global registered list.
- **Event Summary Report:** Final analytical summaries mapping resource allocation levels, total enrollment counts, and performance highlights.
- **Registration Confirmation Messages:** Programmatic triggers providing active feedback loops verifying structural sign-up completions.

## 4. WEBSITE ARCHITECTURE & MODULES

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The platform utilizes a structured modular architecture divided into specific operational touchpoints:

- **Home Page:** Public dashboard introducing recent announcements, current updates, and upcoming institutional events.
- **Student Login/Register Module:** Secure entry interface enabling student authentication profile creations and password recoveries.
- **Event List Interface:** Live catalog dashboard outlining all scheduled events filtered by chronological parameters.
- **Event Details Module:** Comprehensive micro-views detailing target criteria, strict schedules, allocated rooms, and resource listings.

- **Registration Interface:** Transactional gateway enabling students to confirm enrollment into chosen activities in one click.
- **Admin Control Panel:** High-clearance command center allowing designated faculties to add events, review user activities, and manage structural databases.
- **Attendance Management Console:** Specialized workspace dedicated to tracking and recording actual attendance ratios during event execution hours.
- **Reports Section:** Analytics module built for processing, filtering, and exporting performance files, sign-up lists, and structural counts.

## 5. DATABASE SCHEMA DESIGN (MYSQL)

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The backend data layer is systematically structured around normalized relational tables to ensure referential integrity and prevent storage redundancy:

**5.1 Table: Student\_Master**

| Column Name       | Data Type         | Constraint Type  | Functional Scope                                                   |
|-------------------|-------------------|------------------|--------------------------------------------------------------------|
| <b>Student_ID</b> | INT / VARCHAR(20) | PRIMARY KEY      | Unique system identification assigned to a student.                |
| <b>Name</b>       | VARCHAR(100)      | NOT NULL         | Captures the complete legal name of the registered student.        |
| <b>Email</b>      | VARCHAR(50)       | UNIQUE, NOT NULL | Maintains verified personal or institutional mailing contact link. |
| <b>Contact</b>    | VARCHAR(15)       | NOT NULL         | Stores active telephone or mobile communication details.           |

**5.2 Table: Event\_Master**

| Column Name       | Data Type    | Constraint Type       | Functional Scope                                           |
|-------------------|--------------|-----------------------|------------------------------------------------------------|
| <b>Event_ID</b>   | INT          | PRIMARY KEY, AUTO_INC | Unique systemic code allocated per generated event entity. |
| <b>Event_Name</b> | VARCHAR(100) | NOT NULL              | Defines the official structural title naming the event.    |

|              |              |          |                                                                  |
|--------------|--------------|----------|------------------------------------------------------------------|
| <b>Date</b>  | DATE         | NOT NULL | Saves the planned calendar execution date for scheduling.        |
| <b>Venue</b> | VARCHAR(100) | NOT NULL | Identifies specific localized campus infrastructure allocations. |

### 5.3 Table: Registration\_Mapping

| Column Name            | Data Type         | Constraint Type       | Functional Scope                                                |
|------------------------|-------------------|-----------------------|-----------------------------------------------------------------|
| <b>Registration_ID</b> | INT               | PRIMARY KEY, AUTO_INC | Unique tracking sequence for individual registration lines.     |
| <b>Student_ID</b>      | INT / VARCHAR(20) | FOREIGN KEY           | Maps directly back to the matching entry inside Student_Master. |
| <b>Event_ID</b>        | INT               | FOREIGN KEY           | Maps directly back to the matching entry inside Event_Master.   |

## 6. TECHNOLOGY STACK & OPERATIONAL ENVIRONMENT

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### 6.1 Software Environment

- **Operating System:** Windows 10 / 11 Ecosystems (Acts as host system for application runtime environments).
- **Frontend Styling Suite:** HTML (Structuring core layout sheets), CSS (Styling, typography, and theme definitions), JavaScript (Executing validation handling and view transitions).
- **Backend Environment Language:** Python (Server-side application processing architecture handling secure dynamic requests).
- **Database Engine:** MySQL (Relational structured management workspace housing data collections safely).
- **Local Host Environment Suite:** XAMPP (Local apache distribution integrating web services and server engines seamlessly).
- **Source Code Workspace:** Visual Studio Code (Standard modular integrated development environment).
- **Target Client Browser:** Google Chrome (Execution engine rendering front-end elements for review).

## 7. FUTURE SCOPE & CONCLUSION

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The future roadmap of the platform plans to integrate custom mobile application alternatives tailored for Android and iOS systems, alongside real-time ticketing and payment handling facilities for premium programs. Automated text triggers via SMS channels and secure email relays will handle notification delivery, while custom QR code-based validation mechanics will streamline on-site check-ins. Additionally, AI-powered predictive engines will offer intelligent recommendations matching distinct student activity preferences, and full cloud integrations will enhance database reliability and secure system scaling.

In conclusion, the College Event Management System delivers a highly reliable web architecture resolving critical inefficiencies common in manual event planning. By centralizing registration tracks, optimizing communication modules, and replacing paper-dependent processes, the system drastically cuts manual friction. It empowers academic communities to manage activities transparently, protect data integrity, and drive superior operational excellence across institutional workspaces.